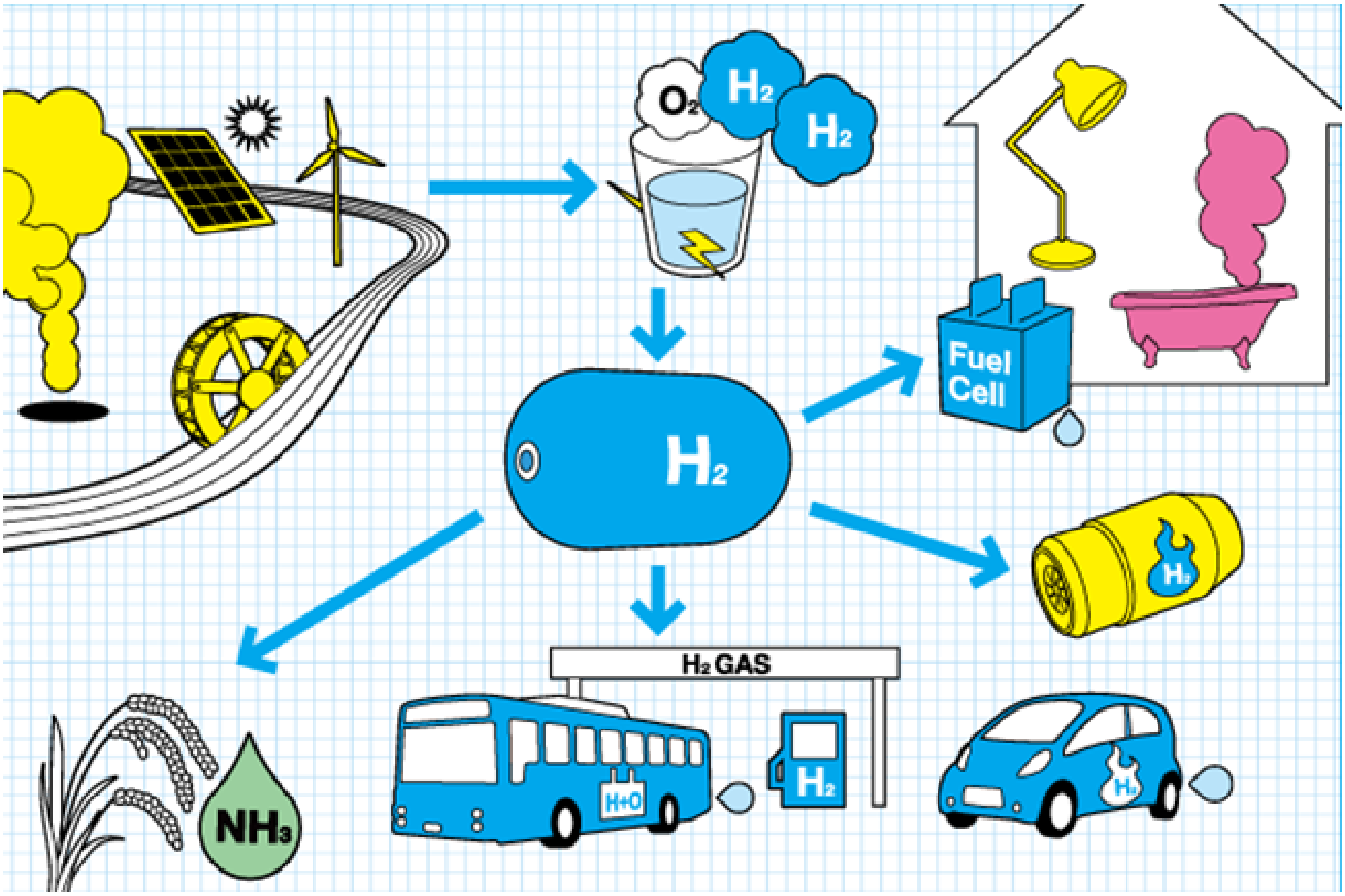
Qualification Pack



Green Hydrogen Plant Entrepreneur

QP Code: SGJ/Q0121

Version: 2.0

NSQF Level: 5

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
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Contents

SGJ/Q0121: Green Hydrogen Plant Entrepreneur

Brief Job Description

Green Hydrogen Plant Entrepreneur would identify the potential market and the client needs/ requirements to propose the right kind of technically and economically feasible solution to set up green hydrogen plant. He/she is also expected to undertake specific works/sub component installations of a green hydrogen plant.

Personal Attributes

This job role requires the individual to be diligent, punctual, attentive and accountable to complete the work at site safely and timely. The individual shall also have an ability to take initiative, handle pressure and shall be motivated with good interpersonal and problem solving skills. The individual shall also be organized with the work and act with integrity while performing multiple tasks for quality and timely. deliverables.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SGJ/N1817: Introduce green hydrogen value chain for entrepreneurs
2. SGJ/N4101: Identify key components of Green Hydrogen Plant and its overall layout
3. SGJ/N1818: Acquire key technical and entrepreneurial skills for green hydrogen production
4. SGJ/N1820: Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production
5. SGJ/N1819: Diversify Micro-entrepreneurship skills in Green Hydrogen
6. SGJ/N0802: Maintain Health & Safety at Green Hydrogen generation project site
7. DGT/VSQ/N0103: Employability Skills (90 Hours)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Entrepreneur

Country	India
NSQF Level	5
Credits	13
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2431.0501 Market Research Associate – Sales & Marketing/Business Development
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) with NA of experience OR 12th grade Pass with 2 Years of experience relevant experience OR Completed 2nd year diploma after 12th with NA of experience OR 10th grade pass with 4 Years of experience relevant experience OR Completed 3 year diploma after 10th with 1 Year of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	24/02/2029
NSQC Approval Date	25/02/2026
Version	2.0
Reference code on NQR	QG-05-GJ-05066-2025-V1-SCGJ
NQR Version	2

Remarks:

Total 480 Hours:(Theory: 155 hours+Practical:145 hours+ 90 hours of employability skills + 90 hours of OJT)

SGJ/N1817: Introduce green hydrogen value chain for entrepreneurs

Description

This NOS provides essential knowledge of the Green Hydrogen sector, its value chain, applications, and entrepreneurial opportunities for setting up green hydrogen enterprises.

Scope

The scope covers the following :

- Fundamentals of hydrogen, green hydrogen processes, applications, and hydrogen economy.
- Entrepreneurial roles and business opportunities across the Green Hydrogen sector.

Elements and Performance Criteria

Introduction to Green Hydrogen

To be competent, the user/individual on the job must be able to:

PC1. Discuss the properties and characteristics of Hydrogen

PC2. Discuss various colour code nomenclature of Hydrogen and role of Green Hydrogen in sustainable energy transition.

PC3. Discuss key aspects related to production, storage and transportation of Green Hydrogen.

PC4. Discuss key aspects of the hydrogen economy

PC5. Discuss the applications of Green hydrogen in industry, transport and power production.

PC6. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain

PC7. Identify opportunities to various renewable energy sources

PC8. Discuss the role and responsibilities of a Green Hydrogen Plant Entrepreneur.

PC9. Demonstrate with chart colour code nomenclature of Hydrogen

PC10. Do an activity for matching the process and source of production of different colour codes of hydrogen

PC11. Draw a flow diagram of green hydrogen production and end uses across the energy system

PC12. Illustrate emerging opportunities for setting up enterprises across Green Hydrogen value chain

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. knowledge of required process in green hydrogen generation

KU2. Green hydrogen domain knowledge

KU3. legislative, organizational, site requirements and technical procedures related to green hydrogen domain

KU4. customer relationship management principles

KU5. renewable energy and green hydrogen market

KU6. key concerns and opportunities in green hydrogen business landscape

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. ability to read and correctly interpret info from different sources- books screens in machines and signage

GS2. understand the various color codes, as per standard electrical, mechanical and civil nomenclature.

GS3. to clearly communicate installation and design instructions to team

GS4. manage relationships with customers

GS5. choose best methods to complete assigned tasks

GS6. apply domain knowledge, observations and data to select course of action to perform tasks related to green hydrogen generation

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Green Hydrogen	**30**	**20**	**-**	**-**
PC1. Discuss the properties and characteristics of Hydrogen	5	-	-	-
PC2. Discuss various colour code nomenclature of Hydrogen and role of Green Hydrogen in sustainable energy transition.	7	-	-	-
PC3. Discuss key aspects related to production, storage and transportation of Green Hydrogen.	3	-	-	-
PC4. Discuss key aspects of the hydrogen economy	3	-	-	-
PC5. Discuss the applications of Green hydrogen in industry, transport and power production.	3	-	-	-
PC6. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain	3	-	-	-
PC7. Identify opportunities to various renewable energy sources	3	-	-	-
PC8. Discuss the role and responsibilities of a Green Hydrogen Plant Entrepreneur.	3	-	-	-
PC9. Demonstrate with chart colour code nomenclature of Hydrogen	-	5	-	-
PC10. Do an activity for matching the process and source of production of different colour codes of hydrogen	-	5	-	-
PC11. Draw a flow diagram of green hydrogen production and end uses across the energy system	-	5	-	-
PC12. Illustrate emerging opportunities for setting up enterprises across Green Hydrogen value chain	-	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1817
NOS Name	Introduce green hydrogen value chain for entrepreneurs
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

SGJ/N4101: Identify key components of Green Hydrogen Plant and its overall layout

Description

This unit outlines the key components of a Green Hydrogen production plant and the overall plant layout

Scope

The scope covers the following :

- Key components of the plant

Elements and Performance Criteria

Components of Green Hydrogen Plant and its layout

To be competent, the user/individual on the job must be able to:

PC1. Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components

PC2. Discuss functions of the key components of electrolyzer stack, renewable power plant, feed water supply unit, gas separator, transformer and rectifier, gas compression unit, etc.

PC3. Explain overall layout of the plant

PC4. Read and interpret various electrical codes, data sheet, safety features etc relevant to the plant.

PC5. Discuss working principles of main components including electrolyzer stack, gas separator, power source, etc.,

PC6. Discuss key technical insights of Green hydrogen system for setting up and running successful enterprises

PC7. Discuss how to perform SWOT analysis

PC8. Discuss safety aspects to be considered while preparing the layout of hydrogen production plant for prevention from fire caused by sparking, e.g. to ensure safe distance of hydrogen from electrical devices like MCBs, switches

PC9. Illustrate the schematic of Green hydrogen production plant

PC10. Illustrate key components of the plant and outline their functions through plant schematic

PC11. Illustrate possible combination of various renewable power sources for generating green hydrogen

PC12. Illustrate how to interpret the Plant Layout including various equipment and material used in a Green hydrogen production facility.

PC13. Demonstrate how to interpret signs, notices and/or cautions at project site

PC14. Illustrate key technical insights of Green hydrogen system for setting up and running successful enterprises

PC15. Perform SWOT analysis for outlining key business opportunities in the value chain

PC16. Show how to perform Proper earthing of mainly the tools to safeguard against static charge

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. to identify the functions of key components of hydrogen generation system and related process
KU2. efficiency, cost and typical specifications, functioning and operating principle of key components
KU3. government and regulatory requirement of green hydrogen business space
KU4. risk associated with green hydrogen business landscape
KU5. legislative, organizational, site requirements and related technical procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. plan and organize work to meet deadlines
GS2. recognize problems and provide solutions
GS3. support the team in scheduling tasks
GS4. analyze critical points in day to day tasks and identify measures to provide solutions
GS5. prepare documentation as per relevant industry standards
GS6. manage relationships with customers and other stakeholders with intent on satisfying its requirements for service delivery

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Components of Green Hydrogen Plant and its layout	**30**	**20**	**-**	**-**
PC1. Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components	2	-	-	-
PC2. Discuss functions of the key components of electrolyzer stack, renewable power plant, feed water supply unit, gas separator, transformer and rectifier, gas compression unit, etc.	4	-	-	-
PC3. Explain overall layout of the plant	4	-	-	-
PC4. Read and interpret various electrical codes, data sheet, safety features etc relevant to the plant.	4	-	-	-
PC5. Discuss working principles of main components including electrolyzer stack, gas separator, power source, etc.,	4	-	-	-
PC6. Discuss key technical insights of Green hydrogen system for setting up and running successful enterprises	4	-	-	-
PC7. Discuss how to perform SWOT analysis	4	-	-	-
PC8. Discuss safety aspects to be considered while preparing the layout of hydrogen production plant for prevention from fire caused by sparking, e.g. to ensure safe distance of hydrogen from electrical devices like MCBs, switches	4	-	-	-
PC9. Illustrate the schematic of Green hydrogen production plant	-	2	-	-
PC10. Illustrate key components of the plant and outline their functions through plant schematic	-	2	-	-
PC11. Illustrate possible combination of various renewable power sources for generating green hydrogen	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Illustrate how to interpret the Plant Layout including various equipment and material used in a Green hydrogen production facility.	-	3	-	-
PC13. Demonstrate how to interpret signs, notices and/or cautions at project site	-	3	-	-
PC14. Illustrate key technical insights of Green hydrogen system for setting up and running successful enterprises	-	3	-	-
PC15. Perform SWOT analysis for outlining key business opportunities in the value chain	-	2	-	-
PC16. Show how to perform Proper earthing of mainly the tools to safeguard against static charge	-	3	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4101
NOS Name	Identify key components of Green Hydrogen Plant and its overall layout
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

SGJ/N1818: Acquire key technical and entrepreneurial skills for green hydrogen production

Description

This unit outlines how to acquire all key technical and entrepreneurial skills required for managing green hydrogen generation

Scope

The scope covers the following :

- Acquiring key skills

Elements and Performance Criteria

Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production

To be competent, the user/individual on the job must be able to:

PC1. Discuss process for setting up hydrogen production plant including required registration, availing low cost financing from Financial Institutions/Venture Capital/Banks, government subsidies, green power, free wheeling/banking of renewable energy etc.

PC2. Discuss various evolving opportunities for setting up of hydrogen production plants at different sites, technologies, services, finance and market ecosystem

PC3. Discuss how to identify the key ingredients of business plan for land, water, renewable energy, finance, technology and market mechanism

PC4. Discuss how to assess the potential and demand of Hydrogen in different sectors i.e. transport, power, building, industry, etc

PC5. Discuss all key aspects of a Detailed Project Report (DPR) and explain how to develop a bankable DPR which includes market demand, storage, technology and finance for hydrogen production

PC6. • Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)

PC7. Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.

PC8. Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.

PC9. Explain different taxation and related compliances to setting up and operations

PC10. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same

PC11. Explain how evolving start-ups and new ventures boost-up hydrogen economy.

PC12. Show how to access required data and analyse key information for performing site survey for installation of Hydrogen generation plant

PC13. Identify different players involved in value chain for hydrogen production for component supply and human resources

PC14. Illustrate all key aspects of a Detailed Project Report (DPR) and show how to develop a bankable DPR for Hydrogen production plant

PC15. Identify various Approvals and clearances requirement

PC16. Demonstrate how system installation cost can vary depending upon the site conditions and related requirements

PC17. Show how O&M related cost for the system can be calculated

PC18. Show how available incentives/subsidy etc. can be availed for setting-up of a Hydrogen production plant

PC19. Show how to Calculate lifecycle cost of Hydrogen production plant along with its business viability including cash flow, annual savings and payback period

PC20. Identify opportunities in using green finance to encourage compliance with green hydrogen certification as per global standards

PC21. Identify opportunities with evolving business models in trading hydrogen as an energy commodity

PC22. Identify new and evolving business opportunities for collaborating with renewable power developers, EPC, system integrators, manufacturers along with downstream application segments

PC23. Explain and Demonstrate how to use relevant reporting and measurement tool for renewable generation/power consumption for hydrogen generation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and SOPs on green hydrogen generation

KU2. document information using appropriate corporate forms

KU3. typical specifications, functioning, operating principle and efficiencies of different types of components for green hydrogen generation

KU4. legislative, organizational, site requirements and technical procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. prepare documentation as per relevant industry standards

GS2. present information in a logical and organized way

GS3. take decision with systematic course of actions and/or response

GS4. read and interpret data from various sources

GS5. apply wide range of factual and theoretical knowledge to select the right course of action to perform tasks related to green hydrogen business landscape

GS6. use acquired knowledge of the process for identifying and handling issues

GS7. follow organisational code of conduct

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production	**40**	**60**	**-**	**-**
PC1. Discuss process for setting up hydrogen production plant including required registration, availing low cost financing from Financial Institutions/Venture Capital/Banks, government subsidies, green power, free wheeling/banking of renewable energy etc.	6	-	-	-
PC2. Discuss various evolving opportunities for setting up of hydrogen production plants at different sites, technologies, services, finance and market ecosystem	4	-	-	-
PC3. Discuss how to identify the key ingredients of business plan for land, water, renewable energy, finance, technology and market mechanism	3	-	-	-
PC4. Discuss how to assess the potential and demand of Hydrogen in different sectors i.e. transport, power, building, industry, etc	4	-	-	-
PC5. Discuss all key aspects of a Detailed Project Report (DPR) and explain how to develop a bankable DPR which includes market demand, storage, technology and finance for hydrogen production	5	-	-	-
PC6. • Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)	3	-	-	-
PC7. Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.	3	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.	3	-	-	-
PC9. Explain different taxation and related compliances to setting up and operations	3	-	-	-
PC10. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same	3	-	-	-
PC11. Explain how evolving start-ups and new ventures boost-up hydrogen economy.	3	-	-	-
PC12. Show how to access required data and analyse key information for performing site survey for installation of Hydrogen generation plant	-	5	-	-
PC13. Identify different players involved in value chain for hydrogen production for component supply and human resources	-	5	-	-
PC14. Illustrate all key aspects of a Detailed Project Report (DPR) and show how to develop a bankable DPR for Hydrogen production plant	-	5	-	-
PC15. Identify various Approvals and clearances requirement	-	5	-	-
PC16. Demonstrate how system installation cost can vary depending upon the site conditions and related requirements	-	5	-	-
PC17. Show how O&M related cost for the system can be calculated	-	5	-	-
PC18. Show how available incentives/subsidy etc. can be availed for setting-up of a Hydrogen production plant	-	5	-	-
PC19. Show how to Calculate lifecycle cost of Hydrogen production plant along with its business viability including cash flow, annual savings and payback period	-	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. Identify opportunities in using green finance to encourage compliance with green hydrogen certification as per global standards	-	5	-	-
PC21. Identify opportunities with evolving business models in trading hydrogen as an energy commodity	-	5	-	-
PC22. Identify new and evolving business opportunities for collaborating with renewable power developers, EPC, system integrators, manufacturers along with downstream application segments	-	5	-	-
PC23. Explain and Demonstrate how to use relevant reporting and measurement tool for renewable generation/power consumption for hydrogen generation.	-	5	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1818
NOS Name	Acquire key technical and entrepreneurial skills for green hydrogen production
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

SGJ/N1820: Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production

Description

This unit outlines how to Oversee the Assembly and storage of Hydrogen generation system along with the O&M of Electrolyzers

Scope

The scope covers the following :

- Oversee process operations

Elements and Performance Criteria

Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production

To be competent, the user/individual on the job must be able to:

PC1. Discuss the Electrolyser Type (PEM, AE, SOEC)
PC2. Explain the technical specification of Electrolyser (PEM, AE and SOEC)
PC3. Explain the Major components of Electrolyser
PC4. Discuss the Operation of PEM and AE and SOEC electrolyser
PC5. Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.
PC6. Discuss the Tools required for Installation, dismantling, removal of components of PEM and AE /Alkaline water electrolysers
PC7. Explain the inputs / outputs of Electrolyzer
PC8. Discuss in detail the input renewable power from various sources and its integration with electrolyser.
PC9. Discuss how to oversee the Installation of Parts and Components of Electrolyser
PC10. Discuss step by step process for Installation of Electrolyser
PC11. Explain the Challenges associated with Hydrogen in storage, handling and transportation
PC12. Discuss how to select and install hydrogen compression and storage system
PC13. Discuss the precautions required to compress and store hydrogen
PC14. Discuss the safety guidelines to be followed as per industry standard
PC15. Demonstrate different types of electrolysers (Pictures, videos,etc.)
PC16. Outline differences in PEM, AE and SOEC Electrolyser
PC17. Draw detailed schematic of different Electrolyzer
PC18. Do activities to consolidate information on procurement cost of different components with make of any type of Electrolyzer along with technical specification sheet

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislative, organizational, site requirements and technical procedures for installation of green hydrogen generation system
KU2. typical specifications, functioning, operating principle and efficiencies of different components of green hydrogen generation system
KU3. key parameters associated with green hydrogen system
KU4. documentation parameters
KU5. identify the installation process and functions of key components
KU6. risks associated with green hydrogen projects

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read warnings, instructions and other text material on product labels, components etc.
GS2. interact with workers and other stakeholders in order to coordinate work processes
GS3. follow code of conduct
GS4. tools & tackles required for installation
GS5. occupational health and safety (OHS) standards and associated risks when working at green hydrogen site
GS6. typical specifications, costs, functioning, installation procedures, operation and maintenance principles, handling procedures and warranties of green hydrogen generation system

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	**20**	**30**	**-**	**-**
PC1. Discuss the Electrolyser Type (PEM, AE, SOEC)	3	-	-	-
PC2. Explain the technical specification of Electrolyser (PEM, AE and SOEC)	2	-	-	-
PC3. Explain the Major components of Electrolyser	1	-	-	-
PC4. Discuss the Operation of PEM and AE and SOEC electrolyser	1	-	-	-
PC5. Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.	2	-	-	-
PC6. Discuss the Tools required for Installation, dismantling, removal of components of PEM and AE /Alkaline water electrolysers	1	-	-	-
PC7. Explain the inputs / outputs of Electrolyzer	3	-	-	-
PC8. Discuss in detail the input renewable power from various sources and its integration with electrolyser.	1	-	-	-
PC9. Discuss how to oversee the Installation of Parts and Components of Electrolyser	1	-	-	-
PC10. Discuss step by step process for Installation of Electrolyser	1	-	-	-
PC11. Explain the Challenges associated with Hydrogen in storage, handling and transportation	1	-	-	-
PC12. Discuss how to select and install hydrogen compression and storage system	1	-	-	-
PC13. Discuss the precautions required to compress and store hydrogen	1	-	-	-
PC14. Discuss the safety guidelines to be followed as per industry standard	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Demonstrate different types of electrolysers (Pictures, videos,etc.)	-	5	-	-
PC16. Outline differences in PEM, AE and SOEC Electrolyser	-	10	-	-
PC17. Draw detailed schematic of different Electrolyzer	-	5	-	-
PC18. Do activities to consolidate information on procurement cost of different components with make of any type of Electrolyzer along with technical specification sheet	-	10	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1820
NOS Name	Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

SGJ/N1819: Diversify Micro-entrepreneurship skills in Green Hydrogen

Description

This unit outlines how to identify entrepreneurial opportunities across Green hydrogen value chain

Scope

The scope covers the following :

- Micro-entrepreneurship

Elements and Performance Criteria

Micro-entrepreneurship opportunities in Green Hydrogen

To be competent, the user/individual on the job must be able to:

PC1. Discuss micro-entrepreneurship opportunities for uninterrupted quality water supply for electrolyzer, water quality check

PC2. Discuss micro-entrepreneurship opportunities for Installation and commissioning of electrolyzer

PC3. Discuss micro-entrepreneurship opportunities for Renewable energy power supply arrangement plant for green hydrogen plant

PC4. Discuss micro-entrepreneurship opportunities for Storage of green hydrogen

PC5. Discuss micro-entrepreneurship opportunities for Installation of pipelines in green hydrogen plant/industry

PC6. Discuss micro-entrepreneurship opportunities for Safe transportation of green hydrogen to consumers

PC7. Discuss micro-entrepreneurship opportunities for Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser

PC8. Discuss micro-entrepreneurship opportunities for Managing safety of the plant

PC9. Discuss micro-entrepreneurship opportunities for Operation and Maintenance of green hydrogen plant

PC10. Discuss micro-entrepreneurship opportunities for Leakage monitoring, purging of hydrogen by CO_2/NH_3 before start of work. Recharging of hydrogen by avoiding air contact by using again CO_2/NH_3 and then filling of hydrogen

PC11. Demonstrate opportunities for each micro-entrepreneurship in the green hydrogen sector

PC12. Identify entrepreneurial opportunities for inspecting the Green hydrogen production system facility, perform equipment’s, product and marketing documentation, quality control, process validation, verify the procedures and records related to the activities of the manufacturing unit etc

PC13. Identify opportunities in labelling/certification for green hydrogen generators and facilitating them comply with relevant standards and regulations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislative, organizational, site and environmental requirements and technical procedures of green hydrogen generation
KU2. different risks associated with the project like cost over-run, project completion risk,etc
KU3. preparation of an activity implementation plan for green hydrogen generation
KU4. safety precautions to be taken
KU5. preparation of preventive maintenance schedule for different components
KU6. customer relationship
KU7. government policies and subsidies for green hydrogen projects
KU8. to identify the functions of key components

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read various color codes, as per standard electrical, mechanical and civil nomenclature
GS2. lead a team or group of technicians and plan and organize service work to meet deadlines
GS3. manage relationships with customers with intent on satisfying its requirements for service delivery
GS4. plan to utilize time and equipment's effectively
GS5. prepare and maintain documentation
GS6. respond appropriately to any queries
GS7. take decision with systematic course of actions and/or response

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Micro-entrepreneurship opportunities in Green Hydrogen	**60**	**40**	**-**	**-**
PC1. Discuss micro-entrepreneurship opportunities for uninterrupted quality water supply for electrolyzer, water quality check	6	-	-	-
PC2. Discuss micro-entrepreneurship opportunities for Installation and commissioning of electrolyzer	6	-	-	-
PC3. Discuss micro-entrepreneurship opportunities for Renewable energy power supply arrangement plant for green hydrogen plant	6	-	-	-
PC4. Discuss micro-entrepreneurship opportunities for Storage of green hydrogen	6	-	-	-
PC5. Discuss micro-entrepreneurship opportunities for Installation of pipelines in green hydrogen plant/industry	6	-	-	-
PC6. Discuss micro-entrepreneurship opportunities for Safe transportation of green hydrogen to consumers	6	-	-	-
PC7. Discuss micro-entrepreneurship opportunities for Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser	6	-	-	-
PC8. Discuss micro-entrepreneurship opportunities for Managing safety of the plant	6	-	-	-
PC9. Discuss micro-entrepreneurship opportunities for Operation and Maintenance of green hydrogen plant	6	-	-	-
PC10. Discuss micro-entrepreneurship opportunities for Leakage monitoring, purging of hydrogen by CO_2/NH_3 before start of work. Recharging of hydrogen by avoiding air contact by using again CO_2/NH_3 and then filling of hydrogen	6	-	-	-
PC11. Demonstrate opportunities for each micro-entrepreneurship in the green hydrogen sector	-	15	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Identify entrepreneurial opportunities for inspecting the Green hydrogen production system facility, perform equipment's, product and marketing documentation, quality control, process validation, verify the procedures and records related to the activities of the manufacturing unit etc	-	15	-	-
PC13. Identify opportunities in labelling/certification for green hydrogen generators and facilitating them comply with relevant standards and regulations	-	10	-	-
NOS Total	**60**	**40**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1819
NOS Name	Diversify Micro-entrepreneurship skills in Green Hydrogen
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

SGJ/N0802: Maintain Health & Safety at Green Hydrogen generation project site

Description

This unit outlines how to perform Health and Safety measures at the Green Hydrogen generation facility

Scope

The scope covers the following :

- Safe work area

Elements and Performance Criteria

Perform Health and safety measures for installing and operating Green hydrogen system

To be competent, the user/individual on the job must be able to:

PC1. Explain the requirements for safe work area at hydrogen generation project site

PC2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems

PC3. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc

PC4. Discuss importance of different detectors and safety tools

PC5. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures

PC6. Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.

PC7. Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing dangerous chemicals

PC8. Explain the importance of administering first aid.

PC9. Identify the personal protective equipment used for the specific purpose

PC10. Identify the hazards associated with hydrogen generation system;

PC11. Identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations

PC12. Discuss all applicable statutory requirements along with safety regulations in terms of fire protection

PC13. Incorporate good housekeeping practices and infection control guidelines.

PC14. Dos and Don’ts during normal operation and maintenance as well as in the event of an accident

PC15. Demonstrate how to administer first aid.

PC16. Demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work.

PC17. Demonstrate the use of fire extinguishers, fire detection and alarm system.

PC18. Show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.

PC19. Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site

PC20. Demonstrate good housekeeping and infection control & prevention practices.

PC21. Show how to perform Periodic mock drills of safety systems to ensure their operation, training of the employees to face accidents and actions to be taken

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. complete documentation work applicable to the role

KU2. work constructively and collaboratively with others

KU3. follow organization's rule-based decision making process

KU4. make timely decisions for efficient utilization of resources

KU5. read various color codes, as per standard electrical, mechanical and civil nomenclature

KU6. read and correctly interpret health and safety instructions and signage

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. obtain authorization from specified officials

GS2. legislative, organization, site requirements and technical procedures

GS3. usage and handling procedure related to safety of hydrogen generation equipment

GS4. importance of wearing protective clothing and other safety gear

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform Health and safety measures for installing and operating Green hydrogen system	**24**	**26**	**-**	**-**
PC1. Explain the requirements for safe work area at hydrogen generation project site	2	-	-	-
PC2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems	3	-	-	-
PC3. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc	1	-	-	-
PC4. Discuss importance of different detectors and safety tools	2	-	-	-
PC5. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	2	-	-	-
PC6. Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.	2	-	-	-
PC7. Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing dangerous chemicals	2	-	-	-
PC8. Explain the importance of administering first aid.	2	-	-	-
PC9. Identify the personal protective equipment used for the specific purpose	2	-	-	-
PC10. Identify the hazards associated with hydrogen generation system;	2	-	-	-
PC11. Identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Discuss all applicable statutory requirements along with safety regulations in terms of fire protection	1	-	-	-
PC13. Incorporate good housekeeping practices and infection control guidelines.	1	-	-	-
PC14. Dos and Don'ts during normal operation and maintenance as well as in the event of an accident	1	-	-	-
PC15. Demonstrate how to administer first aid.	-	3	-	-
PC16. Demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work.	-	5	-	-
PC17. Demonstrate the use of fire extinguishers, fire detection and alarm system.	-	5	-	-
PC18. Show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.	-	4	-	-
PC19. Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site	-	3	-	-
PC20. Demonstrate good housekeeping and infection control & prevention practices.	-	3	-	-
PC21. Show how to perform Periodic mock drills of safety systems to ensure their operation, training of the employees to face accidents and actions to be taken	-	3	-	-
NOS Total	**24**	**26**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0802
NOS Name	Maintain Health & Safety at Green Hydrogen generation project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC5. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC6. recognize the significance of 21st Century Skills for employment

PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC8. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC9. use basic English for everyday conversation in different contexts, in person and over the telephone

PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC11. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC12. identify career goals based on the skills, interests, knowledge, and personal attributes

PC13. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC15. use active listening techniques for effective communication

PC16. communicate in writing using appropriate style and format based on formal or informal requirements

PC17. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC18. communicate and behave appropriately with all genders and PwD

PC19. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC22. identify common components of salary and compute income, expenses, taxes, investments etc

PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC24. operate digital devices and use their features and applications securely and safely

PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC26. display responsible online behaviour while using various social media platforms

PC27. create a personal email account, send and process received messages as per requirement

PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC29. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC33. identify different types of customers and ways to communicate with them

PC34. identify and respond to customer requests and needs in a professional manner

PC35. use appropriate tools to collect customer feedback

PC36. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC37. create a professional Curriculum vitae (Résumé)

PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC39. apply to identified job openings using offline /online methods as per requirement

PC40. answer questions politely, with clarity and confidence, during recruitment and selection

PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. components of salary and how to compute income and expenditure
KU12. importance of maintaining safety and security in offline and online financial transactions
KU13. different legal rights and laws
KU14. different types of digital devices and the procedure to operate them safely and securely
KU15. how to create and operate an e- mail account
KU16. use applications such as word processors, spreadsheets etc.
KU17. how to identify business opportunities
KU18. types and needs of customers
KU19. how to apply for a job and prepare for an interview
KU20. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all to maintain effective work relationship
GS4. how to work in a virtual mode, using various technological platforms
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**3**	**4**	**-**	**-**
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**5**	**-**	**-**
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at

each examination/ training center based on these criteria.

6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N1817.Introduce green hydrogen value chain for entrepreneurs	30	20	0	0	50	11
SGJ/N4101.Identify key components of Green Hydrogen Plant and its overall layout	30	20	0	0	50	11
SGJ/N1818.Acquire key technical and entrepreneurial skills for green hydrogen production	40	60	0	0	100	22
SGJ/N1820.Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	20	30	0	0	50	11
SGJ/N1819.Diversify Micro-entrepreneurship skills in Green Hydrogen	60	40	0	0	100	22
SGJ/N0802.Maintain Health & Safety at Green Hydrogen generation project site	24	26	0	0	50	12
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	11

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	**224**	**226**	**-**	**-**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.